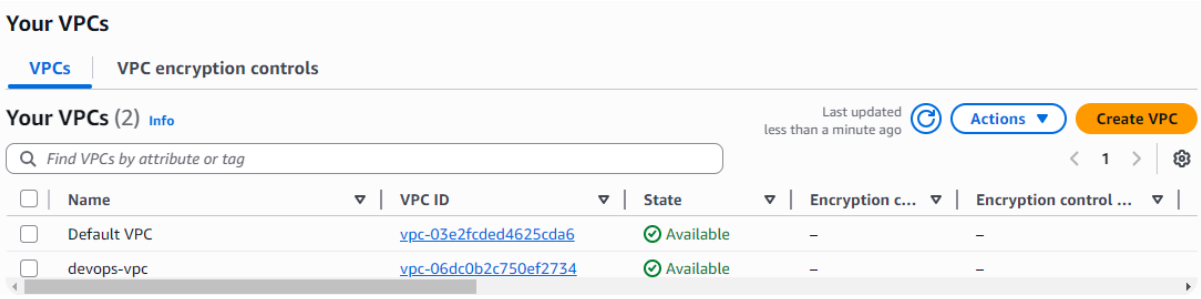
DEPLOYMENT DOCUMENTATION

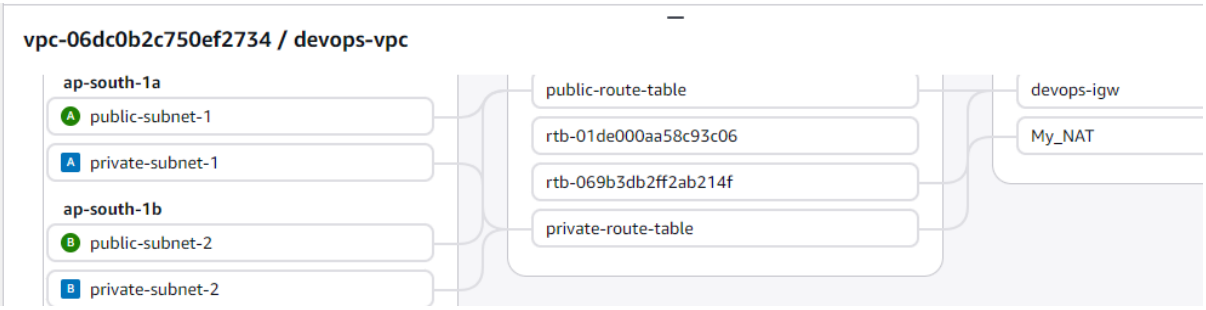
 # Step 1: Login into AWS Account

 # Step 2: Create VPC



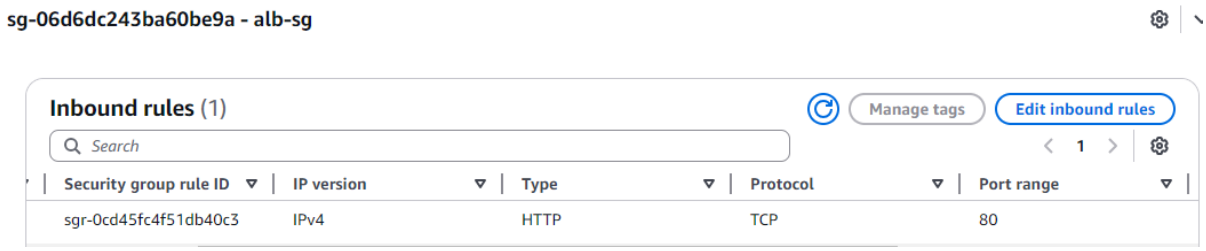
 # Step 3: Create VPC Components

It include Public & Private subnet, Route Tables, Internet Gateway, NAT Gateway



 # Step 4: Create Security Groups

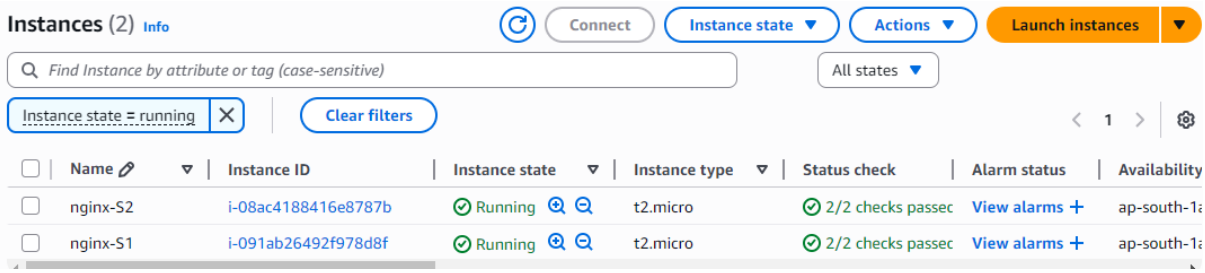
ALB Security Group, EC2 Security Group & Endpoint Security Group



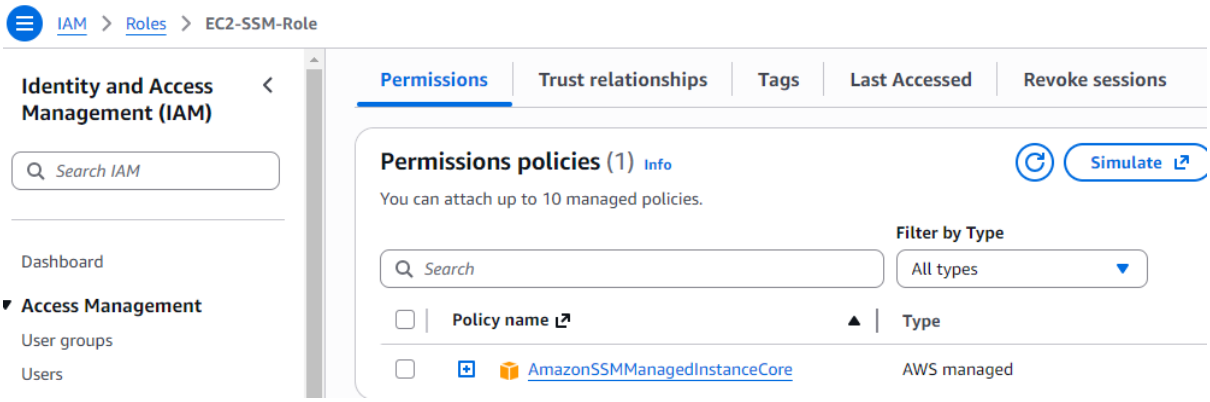
alb-sg ➔ Inbound HTTP 80, Source: 0.0.0.0/0 , Outbound: All traffic

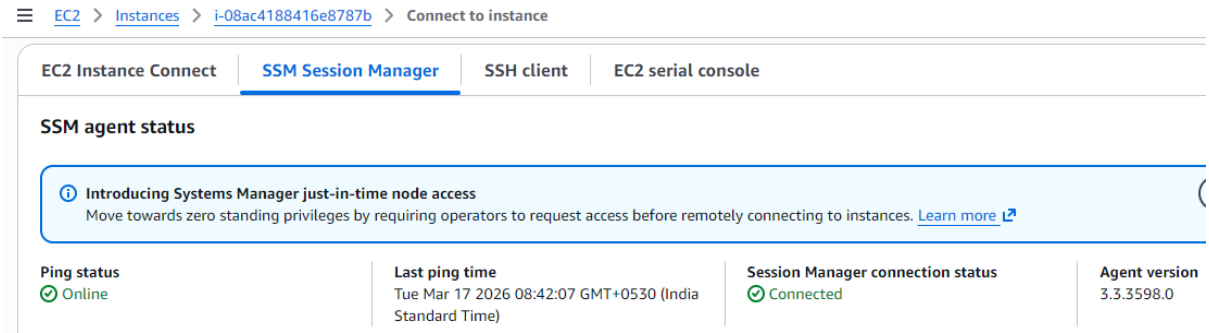
ec2-sg ➔ Inbound HTTP 80, Source: alb-sg, Outbound: All traffic

 # Step 5: Launch Two instance in private Subnets



 # Step 6: Connect these EC2s with AWS System Manager, Attach IAM role to these two instances having AmazonSSMManagedInstanceCore Policy





 # Step 7: Connect EC2, Install & Start Nginx Service in both EC2s

sudo yum update -y
sudo yum install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
sudo systemctl status nginx

```
[ec2-user@ip-10-0-3-120 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-03-16 18:28:57 UTC; 8h ago
    Process: 28940 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 28941 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 28942 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 28943 (nginx)
      Tasks: 2 (limit: 1120)
     Memory: 2.8M
        CPU: 616ms
     CGroup: /system.slice/nginx.service
             ├─28943 "nginx: master process /usr/sbin/nginx"
             └─28944 "nginx: worker process"
```

 # Step 8: Open Nginx index.html file & put the content which you want to display
sudo nano /usr/share/nginx/html/index.html

 # Server 1

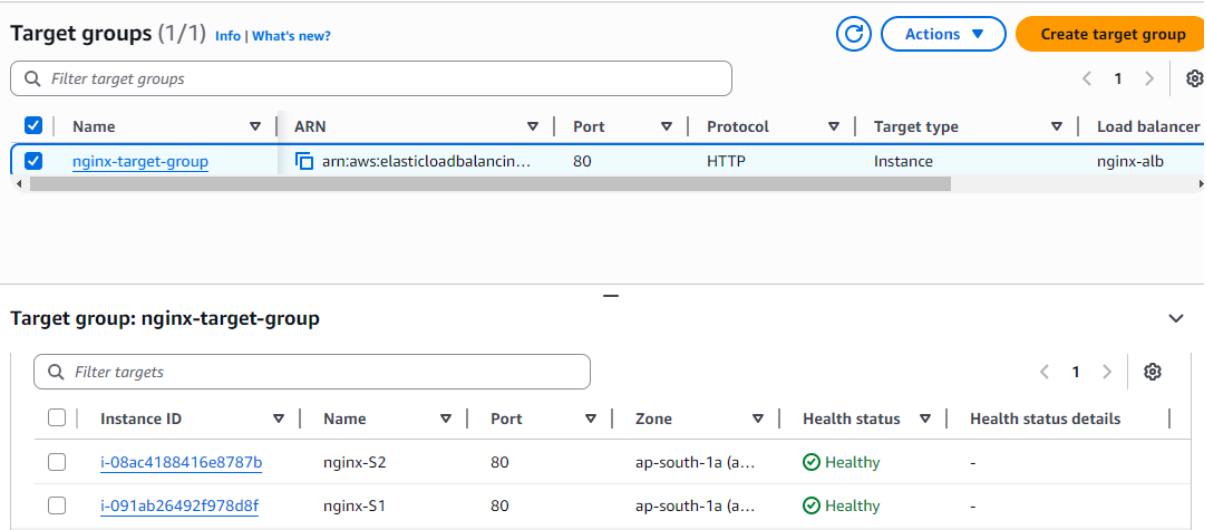
```
[ec2-user@ip-10-0-3-120 ~]$ sudo cat /usr/share/nginx/html/index.html
<!DOCTYPE html>
<html>
<head>
    <title>Hello World</title>
</head>
<body style="text-align: center; font-family: Arial; margin-top: 100px;">
    <h1>Hello World!</h1>
    <p>Instance ID: <strong>i-08ac4188416e8787b</strong></p>
    <p>Availability Zone: <strong>ap-south-1a</strong></p>
    <p>Served by: <strong>Nginx</strong></p>
</body>
</html>
```

 # Server 2

```
[ec2-user@ip-10-0-3-57 ~]$ sudo cat /usr/share/nginx/html/index.html
<!DOCTYPE html>
<html>
<head>
    <title>Hello World</title>
</head>
<body style="text-align: center; font-family: Arial; margin-top: 100px;">
    <h1>Hello World!</h1>
    <p>Instance ID: <strong>i-091ab26492f978d8f</strong></p>
    <p>Availability Zone: <strong>ap-south-1a</strong></p>
    <p>Served by: <strong>Nginx</strong></p>
</body>
</html>
```

 # Step 9: Create Target Groups & Register both Instances

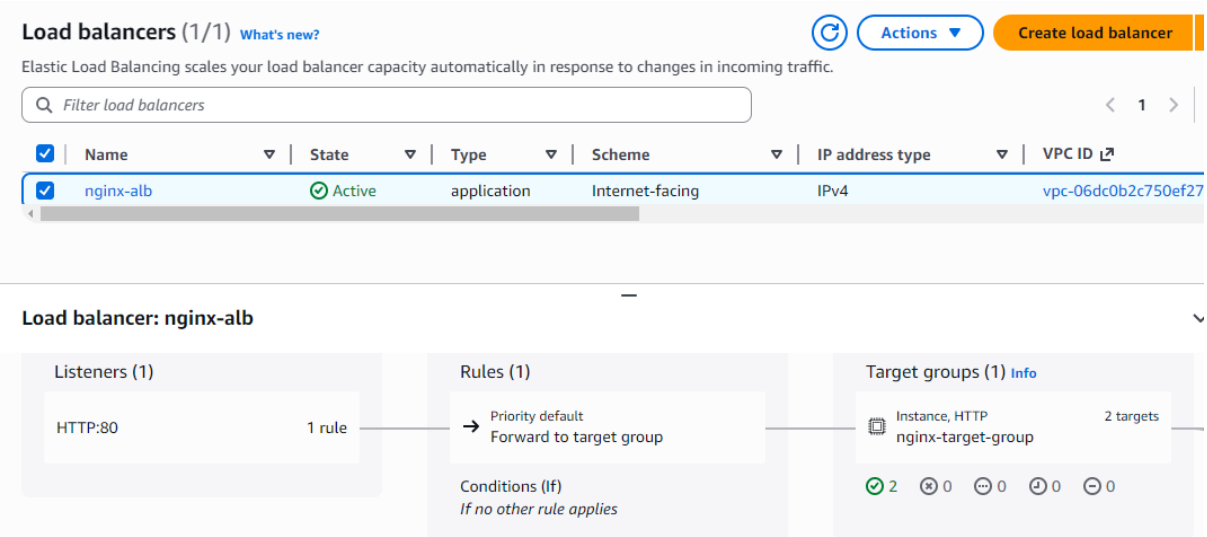
Target Group Configuration ➔
Target Type: Instance
Protocol: HTTP
Port: 80
Name: nginx-target-group
Health Check ➔ Protocal: HTTP, Path: /



 # Step 10: Create Application Load Balancer

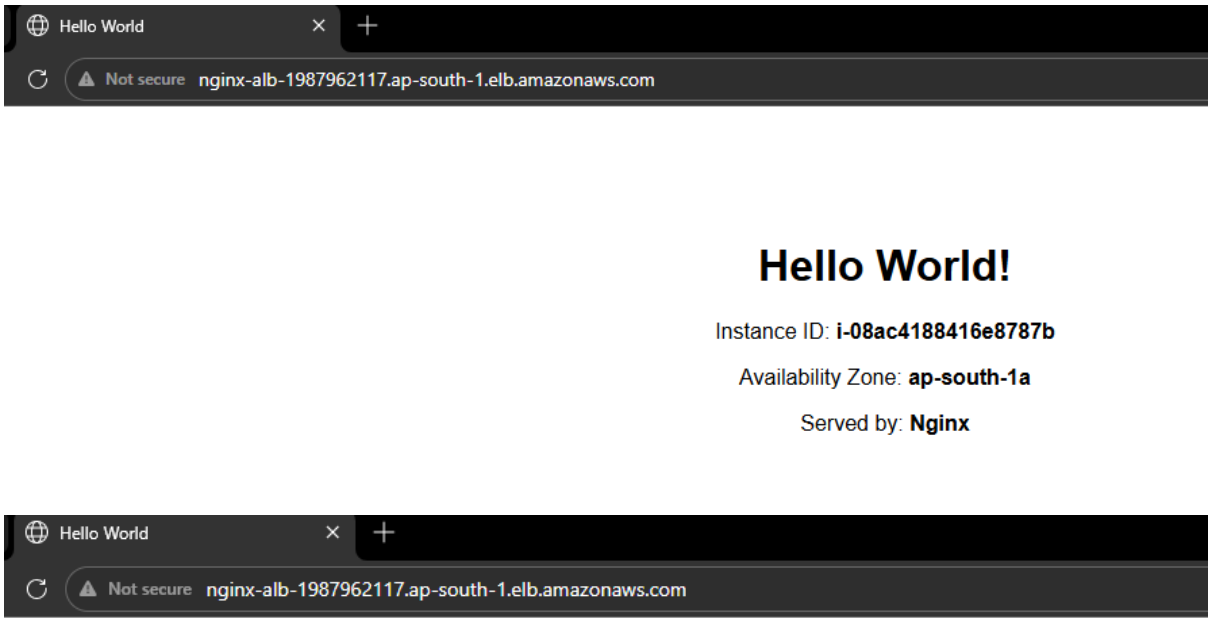
Configuration ➔
Name: nginx-alb
Scheme: Internet Facing
IP type: IPv4
Networking: Select both public subnets
Security Group: alb-sg

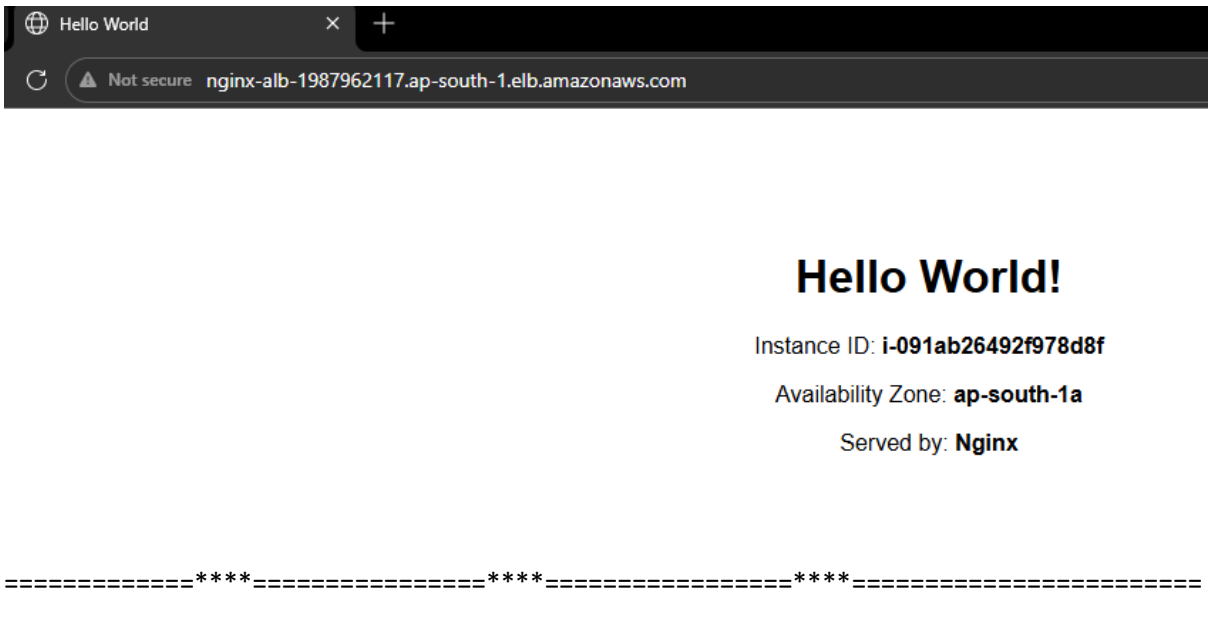
Listener: HTTP 80, Forward to nginx-target-group



 # Step 11: Test Application by hitting ALB DNS, Refresh DNS to see change in static content of nginx

 # Results:


Hello World
Not secure
nginx-alb-1987962117.ap-south-1.elb.amazonaws.com
Hello World!
Instance ID: i-08ac4188416e8787b
Availability Zone: ap-south-1a
Served by: Nginx


Hello World
Not secure
nginx-alb-1987962117.ap-south-1.elb.amazonaws.com
Hello World!
Instance ID: i-091ab26492f978d8f
Availability Zone: ap-south-1a
Served by: Nginx

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